

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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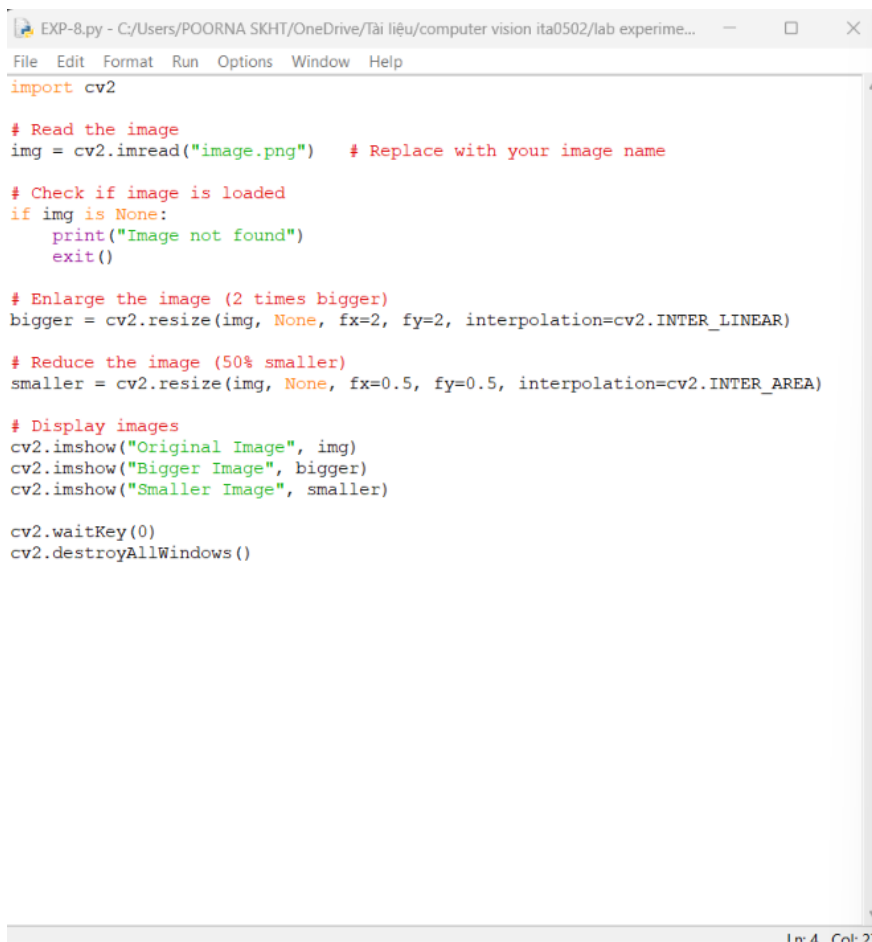
EXPNO:8

AIM: Scaling an image to its Bigger and Smaller sizes.

Algorithm:

- Import the required libraries, mainly cv2, and read the input image using cv2.imread().
- Check the image size using image.shape so you know the original width and height before scaling.
- For **bigger size**, resize the image using scale factors greater than 1, or give a larger width and height using cv2.resize().
- For **smaller size**, resize the image using scale factors less than 1, or give a smaller width and height using cv2.resize().
- Display or save the resized images, using suitable interpolation such as cv2.INTER_CUBIC for enlarging and cv2.

Code:

A screenshot of a Python IDE window titled 'EXP-8.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experime...'. The window has a menu bar with 'File', 'Edit', 'Format', 'Run', 'Options', 'Window', and 'Help'. The code is as follows:

```
import cv2

# Read the image
img = cv2.imread("image.png") # Replace with your image name

# Check if image is loaded
if img is None:
    print("Image not found")
    exit()

# Enlarge the image (2 times bigger)
bigger = cv2.resize(img, None, fx=2, fy=2, interpolation=cv2.INTER_LINEAR)

# Reduce the image (50% smaller)
smaller = cv2.resize(img, None, fx=0.5, fy=0.5, interpolation=cv2.INTER_AREA)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Bigger Image", bigger)
cv2.imshow("Smaller Image", smaller)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

The status bar at the bottom right shows 'Ln 4, Col 27'.

OUTPUT :



RESULT: The image was successfully resized to both larger and smaller dimensions using the OpenCV `resize()` function.